Discovering and Harmonizing Pathogen Genomic Metadata with Large Language Models

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Introduction:

- Genome sequencing has drastically improved public health by enabling the monitoring, characterization, and identification of emerging pathogens.
- Metadata, data that describes data, is essential for supporting accurate analysis, understanding genomic information, and informing public health decision-making
- Advances in LLMs present an opportunity to automate and standardize metadata extraction.
- This project aims to develop an LLM-based pipeline that automates the curation and harmonization of path

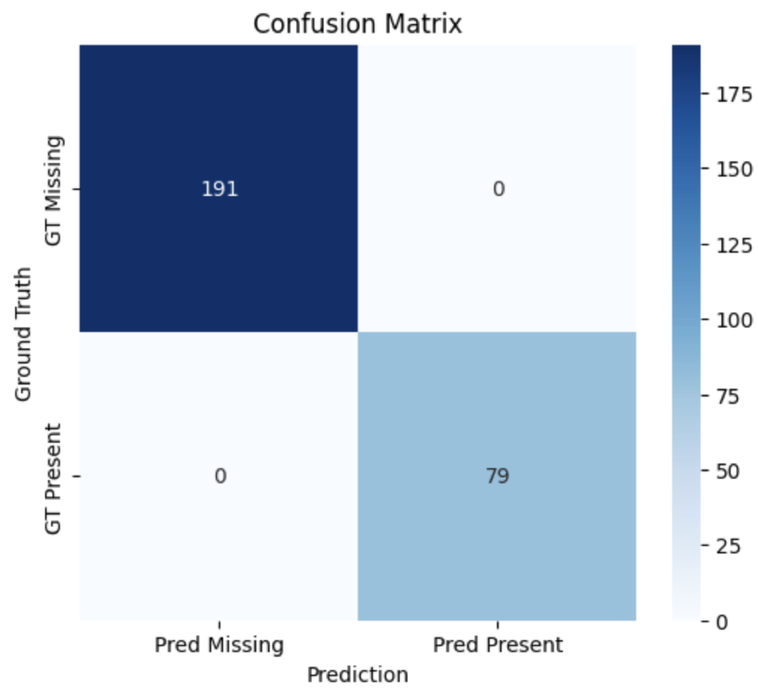
Methods:

1. **Reference dataset**: Contains the BioProject accession IDs, and BioSample accession IDs Organization for each pathogen (SARS-CoV-2, Influenza, Respiratory Syncytial Virus, Tuberculosis, and Candida Auris)
2. **LLM Selection**: Determine the performance metrics of metadata extraction between different LLMs (Gemini, Perplexity, Claude, ChatGPT, and Grok)
3. **Prompt Engineering**: Optimal zero-shot prompt for LLM query
4. **Gold Standard**: Create a dataset of ground truth metadata used for extraction evaluation using standard information retrieval evaluation methods using two independent reviewers and a third to compare results.
5. **Evaluation**: Ensure standardization, quantitative and qualitative performance

Attribute	Coverage (%)
Bioproject Accession	100
Biosample accession	100
collected_by	79.3
collection_date	94
geo_loc_name	99.1
host	95.7
host_disease	72.4
isolation_source	94
lat_lon	30.2
strain	34.5
isolate	0
culture_collection	0
genotype	0
host_age	7.8
host_description	0
host_disease_outcome	0
host_disease_stage	0
host_health_state	0
host_sex	6
host_subject_id	15.5
host_tissue_sampled	0
passage_history	0
pathotype	0
serotype	4.3
serovar	0
specimen_voucher	0
subgroup	0
subtype	8.6

Table 1: Percent coverage of attributes across BioSamples

Implementing LLMs for pathogen metadata extraction can reduce processing time by 98% compared to manual curation.



Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1: 1.0

Fuzzy score: 100.0
Hallucination Rate: 0.0
Time Reduction: 98.1%

Figure 2: Confusion matrix comprised of extracted metadata from one pathogen utilizing a prompting method of uploading the HTML of BioSample webpage.

Results:

Time of Manual vs Automated Metadata Extraction:

Manual extraction: ~ 1842 Seconds (30 Minutes per paper)

Automated extraction:

- ChatGPT: 35 seconds
- Automated extraction resulted in at least a 98.1% reduction in time

Accuracy of Automated Metadata Extraction: 100%

Utilizing the downloaded HTML of the BioSample webpage results in an accuracy of 100% and decreases the time of curation by 98.1%.

Pathogen: clinical or host-associated sample from [Candida] auris

Identifiers
BioSample: SAMN13761842; Sample name: B8441; SRA: SRS5961865

Organism
Candidozyma auris
cellular organisms; Eukaryota; Opisthokonta; Fungi; Dikarya; Ascomycota; saccharomyceta; Saccharomycotina; Pichiomycetes; Serinales; Metschnikowiaceae; Candidozyma

Package
Pathogen: clinical or host-associated; version 1.0

Attributes

isolate	B8441
collected by	Pakistan
collection date	2008-11-05
geographic location	Pakistan
host	Homo sapiens
host disease	Candida infection
isolation source	Burn wound
latitude and longitude	30.3753 N 69.3954 E

BioProject
PRJNA595978 Candidozyma auris
Retrieve all samples from this project

Submission
cdc-ncezid-mdb; 2020-01-08

Accession:
SAMN13761842
ID:
13761842

Figure 1: Example input to ChatGPT (HTML of BioSample Webpage)

Discussion:

While general metadata, such as the geological location and host are consistently available, both host and pathogen sub-classification are excluded. Although the completeness of metadata is rather inconsistent (Table 1). The LLM performed extremely well.

Next steps are to perform evaluation on additional BioProjects along with providing multiple BioProjects and BioSamples at one time to see how the pipeline performs. Additionally, exploring the use of Agentic AI in attempt to streamline fetching the HTML webpage for use in prompting.

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